

Haowen Xu, Ph.D.

CONTACT INFORMATION

Research Fellow (Level B)

School of Built Environment

UNSW Sydney

Kensington, NSW, Australia

Work Email: haowen.xu.phd@gmail.com

Official Email: haowen.xu1@unsw.edu.au

Research Scientist

Computational Sciences and Engineering Division

Oak Ridge National Laboratory (ORNL), U.S. Department of Energy (USDOE)

Oak Ridge, Tennessee, USA

Official Email: xuh4@ornl.gov

Adjunct Associate Professor

Department of Industrial and Systems Engineering

University of Tennessee, Knoxville

Knoxville, Tennessee, USA

WEB PROFILE

[Personal Website](#) – [Google Scholar](#) – [LinkedIn](#) – [ResearchGate](#) – [ORCID](#)

PROFILE

Synopsis of Credentials:

- Six years of post-Ph.D. research experience with:
 - 1 Significant Event Award (SEA) from ORNL in 2020
 - 3 Invention Disclosures (Nonprovisional U.S. Patent) from 2019-2025
 - 2 US Patents in Smart City Applications (Application Number - [US20250131819A1](#) & [US20250128708A1](#))
 - Holds **Permanent Residency in Australia**; eligible to work without visa sponsorship
- 25 peer-reviewed journal publications and over 30 peer-reviewed conference publications, presentations, white papers, and technical reports
 - 8 First/corresponding-author publications in premier journals such as the IEEE Transactions on Intelligent Transportation System (T-ITS), IEEE Transactions on Intelligent Vehicles (T-IV), Science of the Total Environment (STOTEN), Environmental Modelling & Software (EMS), Journal of Environmental Management (JEM), and Journal of Hydrology (JoH).
 - **Over 980 citations** and an **h-index of 17** on [Google Scholar](#).
- Received best research/poster awards from world-renowned professional associations, such as the International Building Performance Simulation Association (IBPSA) and Institute of Electrical and Electronics Engineers (IEEE).
- Co-Principal Investigator (Co-PI) on multi-million-dollar national projects funded by the U.S. Department of Energy's Advanced Research Projects Agency-Energy (ARPA-E) and the Nuclear Safety Research and Development (NSRD) Program.
- Research outcomes have been featured and highlighted by media such as CNN, and professional association, such as the American Water Resources Association (AWRA) Impact and International Association for Hydro-Environment Engineering and Research (IAHR) Hydro-link magazine.
- *Domain expertise* in urban informatics, smart and sustainable cities, urban digital twins, and IoT.
- Serving as an *Adjunct Associate Professor* at the University of Tennessee Knoxville (UTK) and co-advising multiple Ph.D. students.

RESEARCH EXPERTISE

Technical Skills:

- Expertise in **software engineering, design and architecture patterns** (e.g., MVC and MVVM), DevOps (Docker and Kubernetes), and a wide variety of frameworks and software stacks for developing web/mobile/cloud/edge-computing applications and large-scale Cyberinfrastructure (CI).
 1. **Languages:** Python, JavaScript, TypeScript
 2. **Backend Framework:** FastAPI, Django, Flask
 3. **Frontend Framework:** React, React Native (for Mobile Apps), Gradio
 4. **Visualization and Visual Analytics:** D3.js, WebGL, Three.js, VTK.js, Highchart
- Proficient in utilizing and fine-tuning **Large Language Models (LLMs)**, including **ChatGPT APIs, Ollama**, and **Retrieval-Augmented Generation (RAG)** frameworks, with practical experience in **LangChain** and **LangGraph** for agentic tool orchestration, alongside expertise in **relational, graph, and vector databases**.
- Proficient in developing Generative AI models and Agentic AI architecture, including VAEs, Transformers, and Diffusion Models.
- Experienced in multiple Python libraries for data science and deep learning, including Pandas, NumPy, SciPy, Keras, and PyTorch.
- Expertise in prominent programming languages and libraries for developing customized visual encodings, visualizations, and visual analytics methodology for exploring complex scientific data and supporting explainable Artificial Intelligence (AI).
- Expertise in various GIS software and spatial analytics tools, such as ArcGIS, ArcPy, QGIS, and GDAL.
- Expertise in 3D modeling and game engines for developing VR and AR applications, including Blender, AutoDesk Maya, ThreeJS, and Unity 3D.

Application Areas:

- AI agents for smart cities using Large Language Models (LLMs) and Prompt Engineering.
- Generative AI for Urban and Environmental Modeling.
- Intelligent Transportation System (ITS) for smart mobility management.
- Visual Analytics for explorative analysis and decision support.
- Cyberinfrastructure, Digital Twin (DT), and Cyber Physical system (CPS) for developing smart cities, buildings, grids, and mobility systems.
- Virtual Reality (VR) and Augmented Reality (AR) for advanced and immersive scientific visualization.
- Urban informatics for decarbonization and sustainable economy.
- Hydroinformatics and Environmental Informatics for water hazards mitigation.

EDUCATION

Ph.D., Civil Engineering

2015 - 2019

The University of Iowa, Iowa City, IA, USA

Cumulative GPA: 3.90/4.00

Dissertation Title: "Data-Driven Framework for Forecasting Sedimentation at Culverts"

Focus Area: Hydroinformatics and Environmental Informatics

Passed the final thesis defense (on December 5th, 2018)

- Received multiple travel grants and research fellowship from NSF, UCGIS, and CGRER
- UI SETP (the University of Iowa Student Employee of the Year) Nominee

Graduate Certificate in Geoinformatics

2014 to 2016

The University of Iowa, Iowa City, IA

Cumulative GPA: 3.94/4.00

MS, Civil and Environmental Engineering

2013 - 2015

The University of Iowa, Iowa City, IA, USA

Cumulative GPA: 4.00/4.00

Dissertation Title: "Prototype Hydroinformatics-based System for Supporting Decision Making in Culvert Design and Monitoring"

- Received a Full-tuition Scholarship

B.S.E., Civil Engineering

2009 to 2013

The University of Iowa, Iowa City, IA, USA

Cumulative G.P.A.: 3.54/4.00; Dean's List

Thesis Title: "Search Engine for Iowa Watershed Data Information System"

- *Graduated with Honors* in Civil Engineering

**PROFESSIONAL
EXPERIENCE**

UNSW Sydney

March 2025 - Present

Lecturer/Research Fellow (Level B)

School of Built Environment

Oak Ridge National Laboratory

June 2021 – March 2025

Research Scientist, Computational Urban Sciences Group

Computational Sciences and Engineering Division

University of Tennessee, Knoxville

September 2024 - September 2025

Adjunct Associate Professor

Department of Industrial and Systems Engineering (ISE)

Oak Ridge National Laboratory

June 2019 – May 2021

Postdoctoral Research Associate, Computational Urban Sciences Group

Computational Sciences and Engineering Division

University of Iowa

August 2013 – May 2019

Graduate Research Assistant

IIHR – Hydrosience & Engineering, Iowa City, Iowa

Institute of Rural Research and Development (IRRAD)

December 2013 – January 2014

Teaching Assistant & Engineering Surveyor & GIS Specialist

Gurgaon, India

**HONORS AND
AWARDS**

- | | |
|------|---|
| 2024 | Best Research Paper Award from the 51st International Conference on Computers & Industrial Engineering (CIE51), UNSW, Australia |
| 2022 | Best Poster Award from the 19th IEEE International Conference on Mobile Ad-hoc and Smart Systems (MASS) |
| 2022 | Award for the Most Creative Demonstration from IBPSA HackSimBuild 2022 competition in conjunction with Building Performance Analysis Conference and SimBuild 2022 |
| 2022 | Invited presenter at the U.S. National Academy of Sciences (NAS) , sharing research on digital twins and virtual reality applications for smart city development |
| 2021 | Keynote presenter at the 1 st GeoScale Workshop |
| 2020 | Significant Event Award (SEA) – ORNL, USDOE |

Release of Real-time Ctwin v1.1 -- The Chattanooga Digital Twin for Mobility/Transportation

- 2018 Wayne L. Paulson Scholarship Award (1000 USD)
- 2018 Graduate College Post-Comprehensive Research Fellowship (9000 USD)
- 2017 UI SETP (the University of Iowa Student Employee of the Year) Nominee
- 2016 Invited presenter at the headquarters of the U.S. Environmental Protection Agency (USEPA) and the U.S. Army Corps of Engineers (USACE)
- 2016 Best Student Oral Presentation Award from the American Water Resources Association (AWRA)

SELECT PROJECTS

Agentic AI Digital Twin for National-Level Intermodal Freight

2023–2025

Funded by U.S. Department of Energy & Advanced Research Projects Agency-Energy (ARPA-E)

- **Led development** of an *agentic AI digital twin* that automates national freight optimization and decarbonization using Large Language Models (LLMs), retrieval-augmented knowledge graphs, and automated tool use of scientific software and mathematical solvers.
- **Built an LLM-powered chatbot** integrating **Gurobi Optimizer**, **AnyLogic**, and the **Freight Analysis Framework (FAF)** for real-time multimodal logistics simulation across highway, rail, and maritime networks.
- **Implemented the Model Context Protocol (MCP)** to enable autonomous AI tool use for analytics, optimization, and decision-making.
- **Developed reasoning and agent-planning capabilities** through fine-tuned LLMs deployed via *Ollama* and *vLLM*, integrating **RAG (LangChain)** and **function-calling via MCP** with a **Gradio** web interface.
- Delivered interactive dashboards for **carbon-aware route planning** and modal-shift analysis, advancing USDOE sustainability goals.
- **Recognized with Best Research Paper Award (CIE51 2024)** and linked to multiple high-impact publications (*Smart Cities 2024*; *Urban Human Mobility 2025*).
- Portfolio and publications: https://haowenxu-phd.github.io/homepage/portfolio/agentic_ai_dt

Explainable AI for Multivariate Time-Series Analytics

2024–2025

Funded by U.S. Department of Energy & Office of Electricity (OE)

- **Developed a Generative AI framework** integrating **Temporal Fusion Transformers (TFT)** and **Variational Autoencoders (VAEs)** for dimensionality reduction and latent-space analysis of multivariate power-grid time-series data.
- **Created an interactive visual analytics dashboard** to diagnose grid signature events (e.g., transformer failures, arcing, fuse blow, animal disturbance) using reduced latent representations of voltage and current data from PMUs.
- Implemented **latent-space mapping with PCA, t-SNE, and UMAP** to reveal correlations and dependencies among complex temporal patterns, enabling fault classification and anomaly detection.
- Combined quantitative metrics with **Human-in-the-Loop evaluation** to ensure explainability, scalability, and performance of transformer-based models over LSTM and CNN baselines.
- Published in *IEEE Access (2025)*, demonstrating a novel explainable AI framework for grid event diagnosis and energy system resilience. Paper available at: <https://ieeexplore.ieee.org/abstract/document/11141392>
- Software stack used for visual analytics development: React, D3js, PyTorch.
- Portfolio and publications: <https://haowenxu-phd.github.io/homepage/portfolio/gsl>

Voxel Wildfire Simulator – 3D AI-Accelerated Fire Spread Modelling

2025–Present

Funded by UNSW & NSW Fire & Rescue

- Developed a **high-performance 3D wildfire simulation framework** using the Python Taichi **parallel computing framework** to model heat transfer and fire propagation across millions of voxels in urban and wildland–urban interface (WUI) environments.
- Integrated LiDAR and GIS-derived voxel models for **realistic representation of fuel types, terrain, and building geometry** at 0.8–5 m resolution.
- Simulated convection, radiation, and conduction under varying wind and moisture conditions, enabling scenario-based analysis of fire dynamics across complex cityscapes.
- Prototyped a **generative AI surrogate model** based on **transformer architecture**, trained on the physical simulator outputs to predict next-step fire states 10–100× faster than the physical model, enabling near-real-time inference with over 90% accuracy.
- Deployed simulations and AI inference on **UNSW’s Katana HPC system**, supporting emergency response and digital twin integration for Newcastle and Liverpool (NSW).
- Created a web-based 3D visualization tool using React and VTK.js to render and interactively display **millions of 3D voxels** representing **fire spread in complex urban environments**.
- Published in *Fire* (2025) and *ISPRS ICWG III/IV Conference on Geoinformation* (2025)
 1. <https://www.mdpi.com/2571-6255/8/8/293>
 2. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5611510
- Portfolio and publications: https://haowenxu-phd.github.io/homepage/portfolio/bushfire_sim

CTwin – Cloud-Based Digital Twin for Real-Time Traffic Optimization

2019–2023

Funded by U.S. Department of Energy & Office of Energy Efficiency and Renewable Energy (EERE)

- **Led full-stack development of CTwin**, a scalable, **cloud-native digital twin platform** enabling real-time traffic **emulation, optimization, and cyber-physical control** of **transportation infrastructure** for smart mobility systems in Chattanooga, Tennessee.
- **Integrated heterogeneous IoT data sources** (radar, cameras, and signal controllers) into a unified, modular cyber-infrastructure enabling lane-level traffic reconstruction and predictive analytics.
- Implemented **agent-based traffic emulation** using traffic-flow theory and **GPU-accelerated computation** to reconstruct vehicle dynamics and support congestion prediction and mitigation.
- **Deployed adaptive signal control algorithms** that reduced congestion and fuel consumption through closed-loop, data-driven feedback between digital twin and physical infrastructure.
- **Delivered multiscale visualization and analytics dashboards** supporting real-time decision-making from corridor to city-wide levels, enhancing urban mobility efficiency and sustainability using **Angular** (frontend), **Django** and **Spring Boot** (backend), and a **PostgreSQL** database.
- Built a **microservice-based architecture** using open-source frameworks for interoperability, scalability, and maintainability using **Docker and Kubernetes**.
- Project received the **Significant Event Award (SEA)** from the U.S. Department of Energy’s Oak Ridge National Laboratory and was **featured by CNN and U.S. government reports** for its national impact.
- **Published multiple peer-reviewed papers in leading journals, including IEEE Transactions on Intelligent Transportation Systems (2023, 2022) and Expert Systems with Applications (2024).**
- Portfolio and publications for CTwin Project: <https://haowenxu-phd.github.io/homepage/portfolio/ctwin>
- Extent Mobile App Project: https://haowenxu-phd.github.io/homepage/portfolio/mobile_app_smart_speed

More demonstrations of my previous projects and publications are available at:

<https://haowenxu-phd.github.io/homepage/portfolio>